

Progress notes: heating, cooling, hydro and the QDM rebuild

For the meeting with Bryn | code version v17 | June 19, 2026

These are working notes, not a formal write up. The aim is to walk through what changed since the last version, the two real bugs we hit and why, how they are fixed, the actual formulas and sources, and the open questions on QDM versus direct that I want your call on.

1. The one paragraph summary

The carbon and accounting side is correct and checked. The first full run then turned up two real bugs in how we put the future climate onto demand and hydro. Both came down to the same thing, multiplicative quantile mapping used on the wrong kind of variable, and both are now fixed with one standard method from the literature. The shedding numbers from the broken run are not real climate results, they are artifacts of the bugs. After the fix the fake shedding should mostly go away, leaving the real result, which is the thing we actually want to look at.

2. What changed since the version you last saw

2.1 Made the operation faithful to Gøtske (the v15 work)

- Checked our dispatch line by line against their published code, not just the paper.
- Fixed the capacity freeze (we were freezing everything, which silently capped emissions and the fuel supply rate). Now only the costed technologies are frozen, the carbon accounting and the fuel supply stay free, exactly as they do it.
- Fixed load shedding cost, location and sizing to their values (high penalty, only at electricity and heat buses, unlimited so it is a true last resort).
- Direct air capture now gets paid the carbon price for the CO₂ it removes (it was not before).
- Emissions now counted from every source that touches the carbon accounting, with a second independent cross check that the numbers close. They do, to 12 figures.
- Carbon price read straight from the design file (about 468 EUR/tCO₂), not re-derived.

2.2 Redesigned the climate methods (v16, then fixed in v17)

- Dropped the old delta method. Two methods now: QDM and direct (defined below).
- Redesigned heating, cooling and hydro. Cooling was a clean win. Heating and hydro had bugs that v17 fixes.
- Picked the C2E wind turbine that matches the network by specific power (SWT120_3600).
- Confirmed the scenario is SSP4.5 from the project description.

3. The two methods, what they mean

- **QDM (primary).** Keep the network's own 2015 chronology. Reshape each variable to carry only the projected change. This pulls out only the climate change and drops the difference between the two datasets.
- **Direct.** Drop in the C2E future year as it is. This also carries the modelling differences between C2E and the network, so it is the harsher of the two. It is Gøtske's own weather year stress idea, extended to a future year.
- We run both. The gap between them gives us an upper and lower take on the answer.

4. The supply side: QDM on capacity factors (this part works)

For wind, solar and run of river, which are capacity factors bounded in $[0, 1]$, we use multiplicative Quantile Delta Mapping (Cannon et al. 2015). For each value we read off the relative change at its own quantile and apply it to the network's series:

$$\hat{x}(t) = x_{\text{orig}}(t) \cdot \frac{q_{\text{fut}}(\tau_t)}{q_{\text{base}}(\tau_t)}, \quad \tau_t = F_{\text{orig}}(x_{\text{orig}}(t))$$

where q_{fut} and q_{base} are the C2E future and baseline quantile functions and τ_t is the quantile of the value. Result clipped to $[0, 1]$.

Why this is fine for capacity factors. They are bounded, so any instability is clipped harmlessly at 1. And the per quantile detail is what you want for wind and solar, because the change in their extremes is what drives shortfalls. Note also that C2E is already bias corrected against ERA5 at source (Wohland et al. 2025), so what we are taking out here is the difference in how the two pipelines convert weather to power (siting rule, turbine fleet, PV model), not raw model bias.

5. The first bug: heating (QDM blew up)

5.1 The symptom

- The daily heating multiplier swung from 0 to 5, with 20 of 32 countries pinned at the 5x cap on some days.
- A 5x heating multiplier is not physical for a warming climate.
- This created phantom heat demand on scattered days, which the frozen heat infrastructure could not meet, so it shed. The heat pumps then chased the phantom demand and pulled the electricity system down too.

5.2 Why it happened, the real reason

Heating energy is **zero inflated**. Most of the year (all summer, much of spring and autumn) heating is at or near zero. Statistically it is precipitation, not temperature. The multiplicative change factor

$$\Delta(\tau) = \frac{q_{\text{fut}}(\tau)}{q_{\text{base}}(\tau)}$$

is a ratio, and at the low quantiles the bottom $q_{\text{base}}(\tau)$ is near zero, so the ratio is mostly noise and blows up. There is no physical ceiling on heating (unlike a capacity factor), so the divergence survives into the dispatch as real demand.

The deeper reason, which decided the fix. The change factor is indexed by the quantile of the value, but the network (ERA5 based) and C2E (climate model based) heating distributions have different seasonal **support**. C2E barely has a spring heating season while the network has a large one. A per quantile map between two distributions that do not line up is not meaningful. When I first tried the textbook QDM zero fix (see next section), it actually made the spike worse (a multiplier of 24 on a constructed mismatch case), which is what told me per quantile is the wrong frame for this variable.

6. The second bug: Norway hydro (volume not conserved)

6.1 The symptom

- Norway alone shed 36 TWh under QDM, which was 43% of all QDM shedding. Sweden and Finland showed the same at smaller scale.
- Norway is almost pure hydro, so this is hydro electricity.

- Under direct, the same Norway shed only 1.96 TWh. Same data, same reservoir.

6.2 Why it happened, and yes it is the volume limit

Two facts combine.

- **Fact one, the reservoir is volume limited.** Norwegian reservoirs are cyclic: they must end the year where they started. So over a year, total generation cannot exceed total inflow. Storage moves energy in time, it cannot create it. If annual inflow drops by X and demand does not, the gap is unserved energy, spread as a steady trickle all year. That is exactly what we see, about 4 GW short continuously, one long stretch from January to July.
- **Fact two, QDM does not conserve the annual total.** It reshapes quantile by quantile. The new annual volume is not the old volume times a clean ratio, it drifts. And the drift is biased down: the unstable large factors land on near zero winter weeks and add nothing, while the well defined melt weeks (which set the annual volume) get whatever the factor does to them. Net, the annual inflow volume drops.

Put together: QDM quietly removes some of Norway’s annual inflow, the cyclic reservoir turns that straight into year round shedding.

The proof it is the transform, not climate. Direct anchors the inflow to the annual volume change, so it conserves volume by construction. Direct Norway sheds 1.96 TWh, QDM Norway sheds 36 TWh, on identical C2E data. That 18 fold gap is a direct measurement of the volume non conservation. If it were a real climate effect, direct would show it too. It does not. This also tells us the underlying C2E inflow change for Norway is benign.

7. The key realisation: one root cause, two costumes

Heating and hydro inflow are the same kind of variable: zero inflated, non negative, multiplicative. Multiplicative QDM fails on both the same way (unstable low tail, does not conserve the mean). It just shows up differently because the physics differs:

- **Heating** has no upper bound, so the instability becomes phantom demand peaks, and peak limited infrastructure sheds.
- **Hydro** is volume limited by the cyclic reservoir, so the failure to conserve volume becomes steady starvation.

We got lucky on the supply side, the same non conservation happens to wind and solar but is harmless because VRE is not volume constrained and is clipped. Only hydro’s physics made it visible.

8. The fix, with the sources

The literature is clear that zero inflated variables like precipitation (and so heating energy and inflow) need special handling, and there is a named, published method for each part of the problem.

8.1 What we now do for heating and hydro inflow

A smoothed seasonal **change factor** with mean conservation, instead of a per quantile map. Three standard ingredients.

1. Smoothed day of year change factor. Build the smoothed seasonal cycle of C2E baseline and future, then the factor is their ratio:

$$f(d) = \frac{\overline{\text{clim}}_{\text{fut}}(d)}{\overline{\text{clim}}_{\text{base}}(d)}, \quad \overline{\text{clim}}(d) = \text{running mean over a 31 day window of the day-of-year mean}$$

Apply it to the network’s own series, preserving the network’s within day and synoptic shape:

$$\hat{x}(t) = x_{\text{orig}}(t) \cdot f(\text{doy}(t)).$$

The key point: the factor is indexed by **calendar position**, not by the value’s quantile. So every winter day gets the winter factor and every summer day gets the summer factor, consistently. This is meaningful even when the two distributions have different seasonal support, which is exactly where the per quantile map failed. This is a change factor or delta scaling method, the dominant approach in climate impact studies (Lazoglou et al. 2024; it is also what the heating and cooling degree day literature uses).

2. Dry season flooring. Where the baseline climatology is below 5% of its annual maximum, there is essentially no heating or inflow to rescale, so the factor is set to 1 rather than a noise driven ratio. This is the change factor analogue of Singularity Stochastic Removal (Vrac et al. 2016), the standard way to handle the dry tail. A final clip to $[0.2, 3.0]$ is a physical guard.

3. Mean conservation. The smoothed factor fixes the shape, but it does not guarantee the total comes out right, so the last step pins the total. For each window (a season, or the whole year) we rescale the series by a single number so its mean change matches what the C2E model actually projects:

$$\text{factor} = \frac{\text{CCS}_{\text{model}}}{\text{CCS}_{\text{realised}}} = \frac{\overline{\text{fut}} / \overline{\text{base}}}{\hat{\bar{x}} / \overline{x_{\text{orig}}}}$$

What each term is, all averaged over the same window:

- $\overline{\text{base}}$ and $\overline{\text{fut}}$ are the mean of the C2E baseline and future series. Their ratio $\text{CCS}_{\text{model}} = \overline{\text{fut}} / \overline{\text{base}}$ is the change the climate model is telling us to apply (for example 0.88, meaning 12% less heating). CCS just means climate change signal.
- $\overline{x_{\text{orig}}}$ is the mean of the network’s own original series, and $\hat{\bar{x}}$ is the mean after we applied the smoothed factor $f(d)$. Their ratio $\text{CCS}_{\text{realised}} = \hat{\bar{x}} / \overline{x_{\text{orig}}}$ is the change we actually ended up with.
- If the two ratios already agree, the factor is 1 and nothing changes. If the smoothing left us a bit off, the factor corrects it so the realised change equals the model change exactly.

This step is the PresRAT correction (Pierce et al. 2015). We need it because plain multiplicative mapping does not keep the mean, so the totals drift without it. **Where we use which window:** heating uses a **seasonal** window, so the winter, spring, summer and autumn totals each match the model. Hydro inflow uses an **annual** window, so the whole year’s inflow volume matches the model. The annual one is the important one for hydro: the reservoir is cyclic, so the year’s generation is capped by the year’s inflow, and getting that volume right is what stops Norway from shedding.

8.2 What we keep unchanged

- Bounded capacity factors (wind, solar, run of river) stay on the QDM above. They are bounded, clip harmlessly, and per quantile change in the VRE extremes is what you want.

8.3 Did it work

- On a constructed mismatch case (C2E with almost no spring heating, network with plenty), the multiplier now caps at about 1.0 instead of 5 (or 24 with the per quantile zero fix).
- On a snowmelt driven cyclic reservoir, the realised annual volume change now equals the C2E model change to four decimal places, which is what keeps Norway alive.
- Both are locked in as regression tests, plus a full synthetic run of the whole pipeline that solves and shows bounded multipliers in the output.

8.4 Sources, all verified

- Cannon, Sobie, Murdock 2015, Journal of Climate 28, 6938 to 6959. QDM, kept for capacity factors.
- Pierce et al. 2015. PresRAT, the mean conservation correction.
- Vrac et al. 2016. Singularity Stochastic Removal, dry value handling.
- Lazoglou et al. 2024. Change factor methods for impact studies.
- Lehner, Nadeem, Formayer 2023, ASCMO 9, 29 to 44. Shows QDM with mean conservation and dry handling satisfies the three demands (match the historical mean, preserve the change signal, conserve the dry frequency).
- Themeßl et al. 2012 (doi 10.1007/s10584-011-0224-4). Alternative frequency adaptation reference.

9. How cooling works (this was a clean win)

The flaw we found. The network's electricity load is built from historical demand, so today's cooling is already baked into it. The old design added cooling on top, double counting it.

- **The fix.** C2E's cooling comes from a model calibrated on observed demand response, so the C2E **baseline** cooling is an estimate of the cooling already embedded in the historical load. So we subtract that embedded baseline cooling, then add the future cooling.
- **QDM cooling:** add the future cooling as a typical year pattern (a climatology), so the cooling change lands at climatologically sensible hours.
- **Direct cooling:** add the raw C2E future cooling, so a hot spell and its matching low wind hour line up. This is the coincidence effect you wanted for the more interesting load shedding story.
- Checked on the real data: cooling shares come out single digit percent, larger in the south than the north (Italy and Spain around 4%, Germany near zero), no clipping. Sensible.

10. Where each method still has a problem

10.1 Direct

- **Hydro: clean.** Direct anchors inflow to the annual volume, so Norway is fine. This is the control that proved the QDM bug.
- **Heating: a real concern.** Direct transplants C2E's raw heating profile onto infrastructure Gøtske sized for the network's own profile, so it sheds 112 TWh of heat in winter (Italy and Germany). But the network's baseline heating and C2E's baseline heating are both history from two different models, so their shape difference is model discrepancy, not climate change. Direct is importing a model discrepancy and calling it stress.
- **Supply and cooling: legitimate stress.** The capacity factor offset and the cooling coincidence are real effects you want to see.

10.2 QDM

- After v17 the heating blow up and the Norway volume bug are fixed.
- **Honest limitation:** because the factor is smoothed, it will slightly understate any real sharpening of cold extremes. At 2042 this is tiny and safe (it will not invent heat peaks). At 2099 it might matter, and that is the point to come back and use the fuller zero handling, which we are keeping as a backup.

11. The questions I want your call on

Q: Direct heating. Do we keep direct transplanting the raw C2E heating profile (and accept that its heat shedding mixes climate change with model discrepancy, flagged as the deliberately harsher case), or do we apply the same robust change factor to heating in both methods, so direct differs from QDM only on supply and cooling, where the difference is real? My lean is the latter, but it is your call.

Q: The reservoir constraint. Two related points on hydro. (1) The C2E inflow data only covers 11 of the 22 reservoir countries (the large hydro ones); the rest keep their 2015 inflow. Do we document this as a data limitation, or proxy the uncovered countries from their neighbours? (2) Gøtske use a constraint on the reservoir level over the year that we currently do not have the data for, so our reservoirs run with what is called **full foresight**: the optimiser can see the whole year's inflow and demand at once and move water around perfectly, filling and emptying with no penalty as long as it ends the year where it started. That is slightly optimistic, because a real operator does not know the future and keeps a safety buffer, so a real system would shed a little more than ours does. Do we source the ENTSO-E reservoir data to add the constraint, or note this as a known, mildly optimistic deviation?

12. Where things stand and next steps

- Code is at v17, on GitHub, with the fix tested and a guide PDF in the docs folder.
- The next run (2042) should show the fake shedding mostly gone. Once that looks right, run across the years (2042, then 2099, then the full set).
- Then build the system designed directly on C2E and compare, which is the second half of the project.